

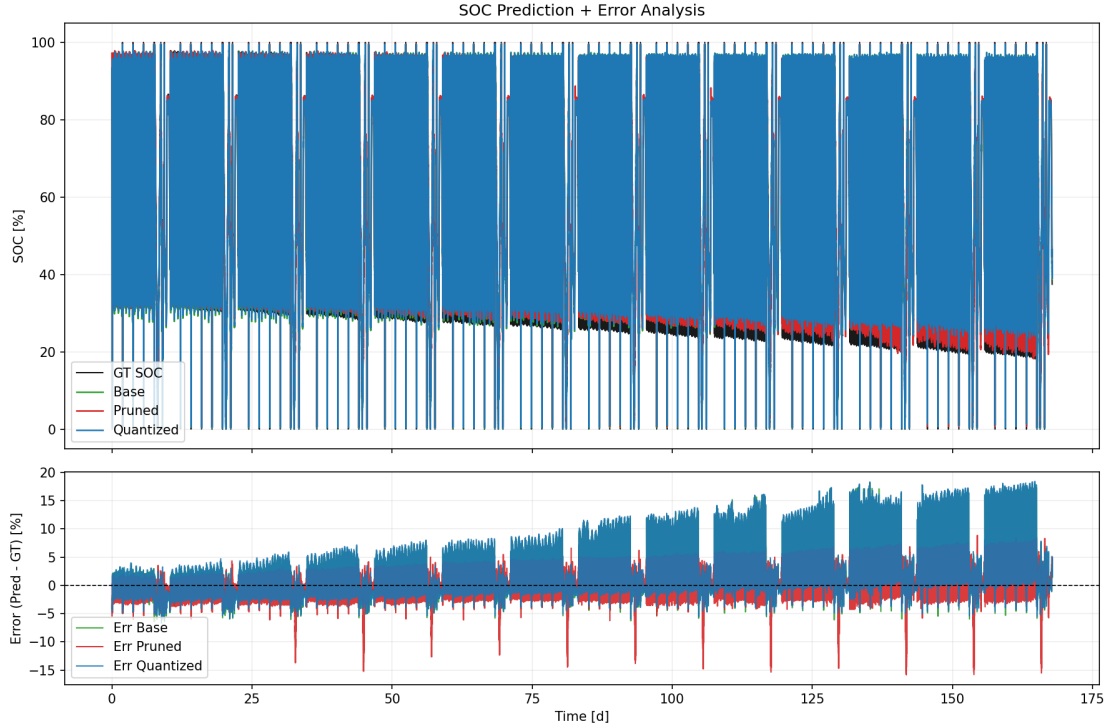


Figure 7.6: Streaming SOC dashboard for the base, pruned, and quantized SOC deployments over the full ageing trajectory. The upper panel compares the three stateful predictions with the SOC reference label, while the lower panel shows the instantaneous prediction error. The near-overlap of the dense and quantized traces is one of the central observations of the chapter.

7.8 Temporal Stability and SOH Filter Interaction

The complete output sequence is divided into ten consecutive parts of equal sample count to distinguish ageing-related changes in task difficulty from a steadily growing compression error. Recurrent states continue across every boundary. For SOC, the MAE of Base rises from 1.116 percentage points in the first part to 4.538 percentage points in the final part. Quantized changes from 1.234 to 4.600 percentage points over the same interval. Despite this common rise in target error, the mean absolute difference between Quantized and Base decreases from 0.332 to 0.271 percentage points. The corresponding Pruned-to-Base difference increases from 0.924 to 2.592 percentage points, even though Pruned has the smallest MAE over the complete trajectory. The SOH segment errors vary with the ageing phase, but neither compressed estimator develops a continuously increasing distance from